

LAB PRACTICE 2

OBJETIVE

Understand Identity and Access Management in Google Cloud. Create cost estimations for Google Cloud Projects.

PREREQUISITES

- Lab teams consist of two students. Each student uses its own Google account. The accounts are referred to as User-A and User-B throughout this document
- Students navigate to console.cloud.google.com and login with their account credentials
- Students must have the Google CLI locally installed

Note: unless otherwise stated, please use the Google Cloud Console to implement the tasks and leave the configuration defaults

Task 1: use predefined IAM roles and grant them to users

1. User-A creates a project called lab-2-a. Go to the IAM & Admin section. Grant the roles to User-B that allows User-B to administer any Compute Engine Instance in the lab-2-a project.

Which are the required predefined IAM roles for these tasks?

2. User-A creates a Compute Engine instance:

- name: `vm2`
- machine type: `e2-medium`
- region: `eu-west-1`
- public image: leave the default `Debian 12`
- network tag: `deep`
- network: `m1-network`

Make sure that the firewall rules for allowing SSH access are in place. Start the virtual machine. Verify that User-A can connect to the Compute Engine instance.

What is the IAM role that is assigned to User-A? What type of role is it (basic/predefined/custom)?

3. User-A creates a Cloud Storage bucket and uploads a file to it.
4. User-B selects the lab-2-a project, inspects the `vm1` configuration and changes the disk size. Then, User-B connects to `vm1`.

Are these actions allowed? Why? Why not?

5. User-B selects the lab-2-a project. User-B lists the Cloud Storage buckets.

Is this action allowed? Why? Why not?

6. User-A grants User-B the additional required roles for completing the previous tasks. Check that now User-B has got access

Which are the required predefined IAM roles for these tasks?

7. Now, User-B wants to create a new bucket in project `lab-2-a`. Rather than asking User-A to grant User-B the required permissions, User-B tries to assign the permission on his own.

What is the result?

8. User-A grants User-B the permissions to be able to create roles and set policies over project `lab-2-a`

Which are the required predefined IAM roles for these tasks?

9. User-B grants itself the permission to create Cloud Storage buckets on `lab-2-a`. Verify that now it works.

What do you think about this method to assign the required permissions? Is it secure?

Task 2: create a custom role and grant it to users

9. In the bucket that we created before we need to have permissions to create, delete and update both buckets and objects. We observe that the `roles/storage.admin` role contains all that permissions. However it also contains the `setIamPolicy` that we do not want to assign. Create a custom role that addresses this situation. Verify that it allows to create objects and does not allow to grant permissions to other user.

Task 3: create and use Service Accounts

10. User-A creates a Service Account named `my-sa`. Grant the service account permission view buckets in the project

11. User-A creates a new Compute Engine instance:

- name: `vm3`
- machine type: `e2-medium`
- region: `eu-west-1`
- public image: leave the default `Debian 12`
- network tag: `deep`
- network: `ml-network` and associates it to the Service Account just created.

12. User-2 connects to `vm3` and tries to list the buckets in the project.

What is the result? Why?

13. User-A adds the missing role to User-B to correct the problem. Verify that now it works as expected.

What is the role that had to be added to User-B

Task 4: assign roles at resource level

14. User-A deletes all roles assigned to User-B at project level

15. User-B assigns Compute Engine Instance administrations right to User-B, only at `vm3` level.

16. User-B tries to change the configuration of **vm3**. User-B tries to create new virtual machines

Is User-B able to do these tasks? Why?

Task 5 (ACTIVITY): IAM best practices

17. Let's assume we work for a company that employs both IT Admins and SW Developers. The IT Admins take care of (amongst other things) to creating, updating and maintaining a set of curated disk images. The SW developers implement SW on new Compute Engine instances by using the images provided by the IT Admins.

Please provide and implement a structure of projects, resources and IAM roles that allows to implement this scenario following the least privilege approach

Task 6: Google Cloud cost estimation

16. Use Google Cloud Price Calculator to estimate what would be the yearly cost of the set up implemented in this lab.
17. Experiment by changing the machine and disks types and configuration and see how the cost varies.

Provide some ideas for reducing costs in the following scenarios:

- The workload only needs to run for 2 hours every day
- The workload needs to run almost non-stop during 3 years
- The workload is interruptible and fault tolerant